

ProofRoad Research Summary

Consolidated research dossier based on ProofRoad v2 plus selected external references. The uploaded proposal remains the primary source for the solution design.

1. Executive summary

ProofRoad is an advisory driver-assistance architecture for mine haul vehicles operating in fog and low visibility. Its core idea is to make speed proportional to road distance that is currently proven clear, while treating unknown information as not clear.

The safety decision is based on a deterministic stopping-distance calculation. Radar is the only sensor allowed to prove the road clear. RGB, thermal and NIR can shorten the proven distance but cannot clear it. Peer/fog-post/gate information can add caution and, only under additional checks, contribute to a more complete proof state.

2. Core formula

Needed distance = speed × reaction time + $\text{speed}^2 / (2 \times \text{braking})$ + buffer.

Braking = tested brake value – slope effect – wet-road penalty.

Safe speed is calculated from the proven distance, reaction time, braking and buffer. If proven distance is not greater than the buffer, safe speed is zero and the truck stops.

The worked example in ProofRoad uses 30 km/h, 1.5 s reaction time, 10 m buffer and 1.4 m/s² braking, producing about 47 m required distance. With 30 m proven, the proposal advises 20 km/h (p. 5).

3. Architecture

Truck safety loop: radar + IMU + route context → SafeCorridor → safe-speed advice → buzzer/vibration/LED/screen.

Assist computer: RGB + thermal + NIR → object naming/fog checks/road strip.

Shared mesh: truck messages, hazard list, gate states and optional signed emergency beacon.

Fog posts: visibility + temperature + humidity + trend.

Ops dashboard: monitoring, analytics and replay only.

4. Key innovation themes

Fresh proof using age counters rather than a shared clock.

Hazards are cleared by actual observation, not by a timer.

Right-of-way at blind curves is derived from remaining room to stop.

Fog posts are audited by passing trucks.

A virtual fleet simulation evaluates throughput against a crawl-in-fog baseline.

Dedicated safety MCU prevents AI/assist-computer failure from disabling the radar warning.

Phantom Obstacle markers let followers see a shared hazard without relying on video.

5. Prototype build priority

Must build: safety MCU with measured braking, radar + RGB, screen + buzzer, alert logic, message age/replay checks, fog-post trend, and failure demonstrations.

Next: thermal, shared hazard list/road strip, blind-curve gates, fleet simulation, camera fog check/NIR, near-miss logging.

Cut first if behind: NIR, 3D dashboard, safety score, worker tags, escort mode, HUD, road-edge LED guides, idle-truck observers, patent wording and absolute safety claims. (ProofRoad p. 12.)

6. Cost planning

ProofRoad estimates approximately ■50,000–■95,000 for the prototype, or approximately ■70,000–■120,000 with one 60 GHz evaluation radar, including two model trucks and the thermal/NIR/driver-interface add-ons. These are planning estimates and must be verified before procurement (p. 11).

7. Safety and evidence boundaries

The uploaded document explicitly says ProofRoad is an engineering proposal, not a safety certification. Radar stand-ins, prototype braking data, artificial mist, two physical trucks, demo keys and limited human-factors testing are all acknowledged limitations.

The proposal says real intervention requires truck-maker approval, functional-safety analysis and mine approval. It references ISO 21815 and ISO 19014 as relevant standards, not as proof of compliance.

8. External research foundation

ISO 21815-1:2022 provides general requirements and guidance for collision warning and avoidance on earth-moving machinery, including detection, operator warnings, automatic intervention and testing; it also states these systems are intended to assist the operator. ■cite■turn0search0■

ISO 19014-1 provides a methodology for determining safety-related control-system performance requirements for earth-moving machinery, while ISO 19014-2 addresses design/evaluation principles for safety-related hardware and architecture. ■cite■turn0search1■turn0search4■

The RADIATE ICRA 2021 paper provides a bad-weather radar perception dataset/reference, useful for adverse-weather radar research but not a substitute for mine-specific validation. ■cite■turn0search6■

EMESRT describes vehicle-interaction improvement as a major mining safety concern and emphasizes the complexity and uncertainty around vehicle-interaction technologies. ■cite■turn0search13■

The Met Office provides a useful technical reference for visibility measurement and meteorological optical range. ■cite■turn0search9■

9. Research conclusion

The strongest research position for ProofRoad is not 'AI solves fog'. It is: the system converts heterogeneous, time-sensitive evidence into a conservative, replayable safe-speed decision; local radar remains the safety floor; other evidence can only add caution; and failures degrade the system toward stop/uncertainty rather than false-clear operation.

The next evidence needed is experimental: mine-representative radar testing, braking characterization, communication tests, sensor calibration, controlled fog/dust tests, fault injection and later driver shadow-mode trials.